Lecture Notes no 12

Blocking in a Factorial Design

In some experiments in which factorial designs are used, because of experimental conditions the experiment cannot be carried out in one time period or one location. In such situations the experiment is done in blocks. Consider the following experiment.

An experiment was conducted to study the life (in hours) of two different brands of batteries in three different devices: radio, camera and portable DVD. The data was collected over a period of two days, each day one complete run of all treatment combinations were observed. The data collected in this experiment is given in table below.

		Day 1			Day 2	
			Device			
Brand of battery	Radio	Camera	DVD	Radio	Camera	DVD
1	8.6	7.9	5.4	8.2	8.4	5.7
2	9.4	8.5	5.8	8.8	8.9	5.9

The model for such an experiment is given by:

$$Y_{ijk} = \mu + A_i + B_j + (AB)_{ij} + R_k + \varepsilon_{ijk},$$

where A_i, B_j,and $(AB)_{ij}$ represent the effect of the ith level of factor A, jth level of factor B and the effect of the $ijth$ level of the AB interaction, respectively. R_k represent the effect of the kth block and ε_{ijk} is the random error, where $i = 1, 2, ..., a$, $j = 1, 2, ..., b$, $k = 1, 2, ..., n$. The ANOVA table for this model, when the Block factor is random, is given in Table 5.20 on page 188.

For the case when the Block factor is fixed, as it is in the experiment described above, the expected mean square for the Block factor is given by:

$$E(MS_R) = \sigma^2 + \frac{ab\sum_{k=1}^{n} R_k^2}{n-1}.$$

The test statistics for testing

$$H_{01} : A_1 = \cdots = A_a = 0,$$

$$H_{02} : B_1 = \cdots\cdot = B_b = 0,$$

and

$$H_{03} : (AB)_{ij} = 0, \text{ for all } i \text{ and } j$$

are

$$F_1 = \frac{MS_A}{MS_E},$$

$$F_2 = \frac{MS_B}{MS_E},$$

and

$$F_3 = \frac{MS_{AB}}{MS_E},$$

respectively.